Q1.

A student investigates the infrared radiation being emitted by different coloured surfaces to the surroundings.

(a) Draw a ring around each correct answer to complete the sentences.

(i) All objects emit and | absorb / conduct / insulate | infrared radiation.

(1)

(ii) Compared with cooler objects, hotter objects emit | less / the same amount of / more | infrared radiation.

(1)

(b) The student pours 300 cm^3 of hot water into each of 3 metal cubes and seals the top of each cube.

(i) Draw a ring around the correct answer to complete the sentence.

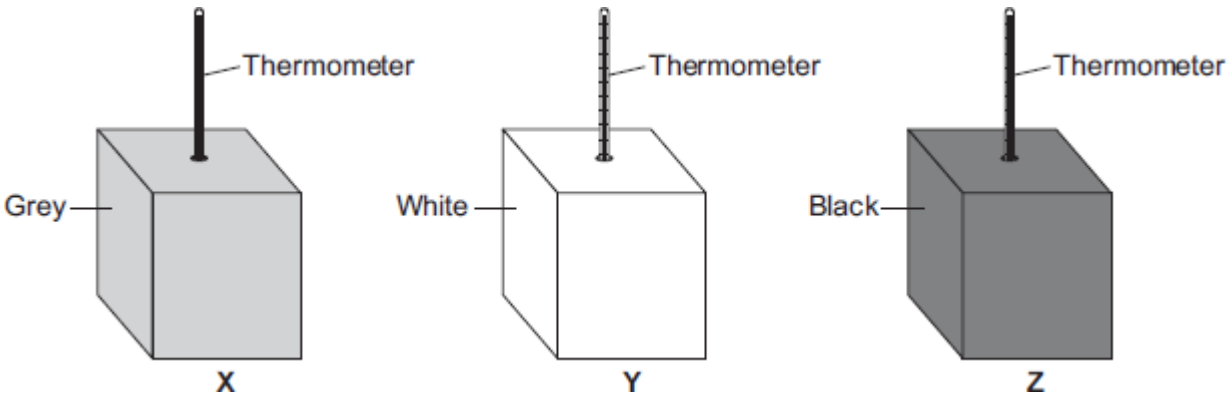
Energy is transferred through the sides of the metal cubes by | conduction. / convection. / radiation.

(1)

Each cube has the same volume.

Each cube is a different colour.

The temperature of each cube is recorded over 10 minutes.

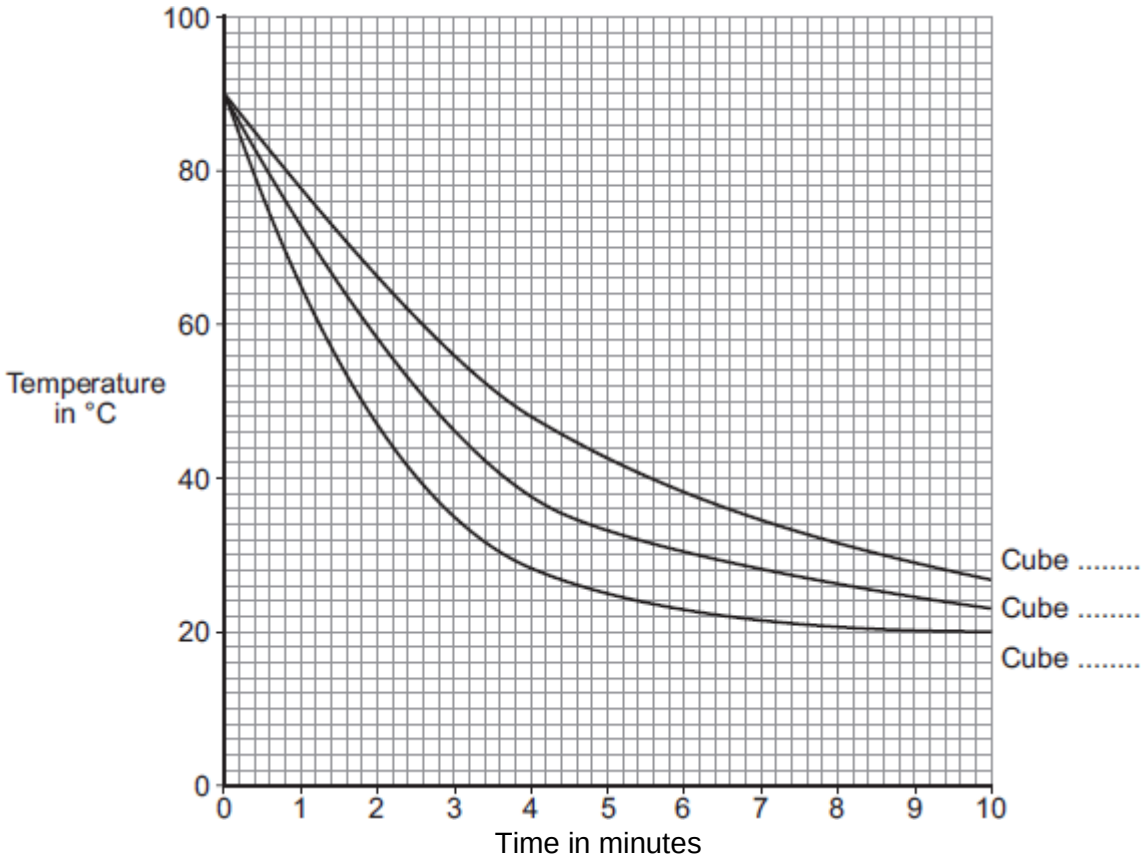


(ii) What is the independent variable in the investigation?

_____

(1)

The results of the investigation are shown on the graph.



(iii) Write the correct letter for each cube, **X**, **Y** or **Z**, next to the lines shown on the graph.

(2)

(iv) All three cubes had the same starting temperature. This was important in the investigation.

Suggest why.

_____

_____

(1)

(v) Some variables are kept the same in an investigation.

What name is given to these variables?

_____

(1)

(Total 8 marks)

Q1.

(a) (i) absorb

1

(ii) more

1

(b) (i) conduction

1

(ii) colour (of cube)

allow colour (of box)

1

(iii) Cube Y

*all three in correct order for **2** marks*

Cube X

Cube Z

*one or two correct for **1** mark*
allow grey for X
allow white for Y
allow black for Z

2

(iv) results can be compared

accept start temperature affects rate of energy transfer
or
start temperature affects how quickly the cube cools down
ignore fair test / reliability / accuracy / control variable / valid

1

(v) control (variable)

allow controls / controlled

1

[8]